

CHAPTER-1

Compiler

To check syntax

Translate java source code (.Java files) -> intermediate code (.Class file)

Interpreter

It shows error

Run Java byte code on any platform

JIT -> help interpreter to run n number of work

JDK -> Java Development kit

Full software package needed to develop Java application

Compiler(Javac)

Debugger(Jdb)- Fix and find Error

Documentational tool (Javadoc)- Generate doc from code

JVM -> Java virtual Machine

Engine that run Java program

[Compile + Run]

JRE -> Java Run time Environment

Execute Java byte code

has core libraries – Java.util, Java.io, Java.lang

Provide the environment where Java application run

IDE -> Interpreted Development Environment

Software Application that provide complete environment to write, edit, compile, debug, and run code easily.

Popular IDE for Java

Eclipse

IntelliJ

Features of IDE:

Code Editor

Compiler/Interpreter Integration

Debugger

Project Management

Other Tools

Magic Byte code

For secured code

It convert Java code → class code

First thing to use [CAFEBABE] is a secret signature at a start of every .Java class file

JIT → Just In Time Compiler/ Interpreter

It is faster

JVM is bit slow

For Example: If you have 1000 times loop, JVM interpret each line by line (slow) but JIT convert all loop into machine code, run into super fast.

Native Machine Code → 0ad binary code

View/read by machine only (machine code)

First should be in capital to save and run Java code.